RIDEWISE - Customer Analytics and Churn Prediction

Overview

1: RideWise Company Profile: RideWise is a leading European mobility technology company serving over 200,000 customers and powering more than 800,000 monthly trips across major cities.

2: Industry Context: RideWise operates in the fast-paced European mobility and transportation technology sector, providing innovative solutions to address the evolving urban transit needs.

3: Current Scale & Market Position: With a strong operational footprint and rapidly growing customer base, RideWise is well-positioned as a market leader in the European mobility technology landscape.

Company Overview

RideWise is a mobility technology initiative supporting European ride-hailing operations with advanced customer analytics and retention systems. The fictional RideWise operates in 5 major European cities (London, Berlin, Amsterdam, Barcelona, and Milan), processing over 200,000 active customers and 800,000+ monthly trips. RideWise faces ***significant challenges with customer retention (25% quarterly churn rate***) and ***ineffective promotion*** strategies. The company's mission is to 1) **Optimize customer lifetime value, while 2) minimizing churn, through 3) data-driven behavioral insights, and 4) predictive customer analytics**.

Business Challenge

- Current customer retention strategies result in ***25% quarterly churn*** among regular users
- *Limited understanding of* ***customer behavioral patterns*** and ***segment differentiation***
- *Promotion campaigns have low uptake rates with* ***unclear ROI measurement***
- *No* ***predictive capability to identify at-risk customers*** *before they churn*
- Manual customer analysis takes weeks per campaign, limiting business agility
- *Lack of* ***real-time customer insights for operational decision-making***
- *No integrated* ***customer analytics*** *system for portfolio-level performance tracking*
- Competitive pressure requires *data-driven customer acquisition and retention strategies*

Project rationale

Mobility companies worldwide face increasing pressure to balance growth with customer retention while optimizing marketing spend and improving operational efficiency in highly competitive markets. This project focuses on:

- Building ***machine learning models*** for customer churn prediction
- Creating customer segmentation using ***clustering algorithms***
- Implementing ***feature engineering pipelines*** for behavioral data
- Developing ***model interpretability*** frameworks for business insights
- Demonstrating ***production-ready deployment*** with ***API integration***
- Integrating business rules and operational constraints into ***model decisions***
- Building ***real-time analytics*** capabilities for customer management

This project equips data scientists with skills in **customer analytics**, **behavioral modeling**, and **production ML deployment** for mobility applications.

Project Objectives

- **Develop** customer churn **prediction models** using **machine learning algorithms**
- **Build customer segmentation system** using **clustering techniques**
- **Implement feature engineering pipeline** for **customer behavioral data**
- **Create** model **interpretability framework** using business-friendly explanations
- **Deploy real-time scoring** capabilities for customer risk assessment
- **Build customer analytics dashboard** and reporting framework
- **Demonstrate API development methodology** for model deployment
- **Create production-ready system** with basic **monitoring capabilities**

Comprehensive Dataset

Table 1: Customers

- Columns: customer_id, signup_date, city, age, registration_source, loyalty_status, account_status, last_activity_date

Table 2: Trips

- Columns: trip_id, customer_id, driver_id, trip_date, pickup_location, dropoff_location, distance_km, duration_minutes, fare_amount, tip_amount, rating_customer, payment_method

Table 3: Customer_behavior

- Columns: customer_id, month_year, total_trips, total_spent, avg_trip_distance, avg_trip_duration, peak_hour_usage, weekend_usage_ratio, cancellation_rate

Table 4: Promotions

- Columns: promo_id, customer_id, promo_type, discount_amount, start_date, end_date, usage_flag, created_date

Table 5: External_factors

- Columns: date, city, weather_condition, is_holiday, local_events, competitor_pricing_index

Dataset Specifications:

- 200,000+ customers across 12 months with activity patterns
- 800,000+ trip records with detailed journey and payment information
- Customer behavioral metrics aggregated monthly for trend analysis
- Promotional campaign data with usage tracking and effectiveness measurement
- External factors data including weather and local events for contextual analysis
- Synthetic data ensuring privacy protection and realistic business scenarios
- Balanced representation across cities, customer segments, and usage patterns

Data Model:

- One-to-many: Each customer may have multiple trips and promotional interactions
- **Behavioral data** LINKS to **customer profiles** for **comprehensive analytics**
- **Time-series structure** SUPPORTS **churn prediction** and **trend analysis**
- **Trip patterns** ENABLE **segmentation** and **usage behavior modeling**
- **External factors** PROVIDE **contextual features** for **improved predictions**

Click the link [Ridewise Dataset] to download the dataset

Technology Stack

Category	Tools/Services Used
Machine Learning	scikit-learn, pandas, NumPy
Data Processing	pandas, NumPy, basic data cleaning
Feature Engineering	Custom pipelines, **RFM analysis**, **behavioral metrics**
Model Interpretation	Basic **feature importance (SHAP)**, business rule explanations
Visualization	matplotlib, seaborn, Streamlit
Model Tracking	**MLFlow**
Backend	FastAPI for model serving
Monitoring	Basic logging, simple performance tracking
Deployment	Local deployment/streamlit deployment
Analytics Libraries	Standard Python data science stack

Model Selection Strategy:

- Customer Segmentation: **K-means clustering for behavioral groupings**
- Churn Prediction: **Logistic Regression and Random Forest for interpretability**
- Feature Engineering: **RFM analysis, usage patterns, temporal features**
- Validation: Time-based splits, **cross-validation**, business sense-checking
- Simple Methods: **Focus on interpretable models** for business understanding

Project scope and phases

Week 1 (June 16-23): Data Analysis & Feature Engineering

- Comprehensive exploratory data analysis of customer and trip data
- Statistical analysis of churn patterns and customer behaviors
- Feature engineering pipeline: RFM metrics, usage patterns, behavioral indicators
- Data quality assessment and customer segmentation foundation

Week 2 (June 23-30): Modeling & API Development

- Build customer segmentation using K-means clustering
- Develop churn prediction models using Logistic Regression and Random Forest
- Implement model validation with time-based splits
- Begin API development with FastAPI for real-time scoring

Week 3 (July 1-7): Integration & Deployment

- Complete API development with core endpoints
- Build analytics dashboard with customer insights and key metrics
- Model performance evaluation and business interpretation
- Final testing, documentation, and presentation preparation

Technical Challenges and Solutions

Data Quality and Consistency:

- Challenge: Customer behavior data may have inconsistencies and missing values
- Solution: Robust data cleaning, validation checks, simple imputation strategies

Class Imbalance in Churn Prediction:

- Challenge: Churn events are less frequent than active customers
- Solution: Balanced sampling, appropriate metrics (precision/recall), threshold optimization

Model Interpretability for Business Users:

- Challenge: Business teams need understandable explanations for customer insights
- Solution: Simple feature importance, business-friendly segment descriptions, clear visualizations

Real-time Scoring Requirements:

- Challenge: Customer risk scores need to be available for operational decisions
- Solution: Lightweight models, efficient API design, basic caching strategies

Success Metrics

Focus Area	Metric
Customer Segmentation	**≥3 distinct segments with clear business differentiation**
Churn Prediction Accuracy	**AUC > 0.80**, **identify top 15% at-risk customers**
Processing Latency	< 1 second for real-time customer scoring
Business Insights Quality	Clear, actionable recommendations for each segment
API Performance	99% uptime, functional core endpoints
Dashboard Usability	5 key metrics, easy to understand visualizations

Evaluation Framework

Statistical Validation:

- Time-based validation with train/test splits
- Cross-validation for model stability assessment
- Business sense-checking of customer segments
- Simple performance metrics relevant to business goals

Business Impact Validation:

- Customer segment analysis with clear business characteristics
- Churn prediction accuracy on recent customer data
- Actionable insights for marketing and retention strategies
- Dashboard usability for business stakeholders

Technical Performance Validation:

- API functionality testing with sample customer data
- Basic load testing for expected usage volumes
- Integration testing between models and dashboard
- Code quality and documentation review

Monitoring and Logging

- Basic model performance tracking with key business metrics
- Simple data quality checks for customer and trip data
- Customer segment stability monitoring over time
- API usage logging with basic performance metrics
- Dashboard usage tracking and user feedback collection
- Monthly business impact assessment with churn rate tracking
- Simple alerting for API downtime or data quality issues
- Documentation updates and model version tracking

Deliverables

- Customer Segmentation Model with K-means clustering and business interpretation
- Churn Prediction System with Logistic Regression and Random Forest models
- Feature Engineering Pipeline with RFM analysis and behavioral metrics calculation
- Model Interpretation Framework with business-friendly explanations and insights
- Real-time Scoring API (FastAPI) with customer risk assessment endpoints
- Customer Analytics Dashboard with segment performance and churn insights
- Basic Monitoring System with model performance and API health tracking
- Technical Documentation including setup guides and API documentation
- Business Insights Report with actionable recommendations for customer strategy
- GitHub Repository with complete codebase, tests, and deployment instructions
- Executive Presentation with business stakeholder recommendations and ROI analysis

Learning Outcomes

- Customer Analytics: Segmentation techniques, churn prediction, behavioral analysis
- Machine Learning: Classification algorithms, clustering methods, model validation
- Feature Engineering: RFM analysis, temporal features, behavioral metrics
- Model Interpretation: Business-friendly explanations, actionable insights
- API Development: Real-time scoring, FastAPI framework, basic deployment
- Data Visualization: Customer dashboards, business reporting, insight communication
- Business Analytics: Customer lifetime value concepts, retention strategies, ROI measurement
- Project Management: End-to-end ML project delivery, stakeholder communication

Suggested Workplan

Step 1: Customer Analytics Fundamentals & Exploratory Data Analysis

Step 2: Feature Engineering for Customer Behavioral Data

Step 3: Customer Segmentation & Clustering Analysis

Step 4: Churn Prediction Models & Customer Risk Assessment